



From Sensors to Scientific Computing: Integrating Science Gateways, HPC, and AI into an Entry-Level IoT Course



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Dr. Kristine Christensen · Moraine Valley Community College — LAN 120: IoT Fundamentals — FacultyHack@Gateways 2026

ABSTRACT

This project redesigns **LAN 120: IoT Fundamentals I**, an entry-level community college course with no prerequisites, to connect hands-on IoT learning with Science Gateways, high-performance computing, artificial intelligence, and data-centered inquiry. Students in the course already work with electronics, Arduino programming, sensors, networking, privacy, and cybersecurity. The redesign extends that foundation by helping students understand what happens to sensor data after it is collected and how similar workflows are used in larger scientific, industrial, and critical infrastructure environments.

Students are introduced to **Science Gateways**, **Jetstream2**, **Jupyter notebooks**, **public datasets**, **cloud computing**, and **HPC resources** through scaffolded activities designed for learners with no prior Python or HPC experience. Using authentic IoT, industrial, and cybersecurity datasets, students run prepared notebook cells, modify selected variables, create visualizations, investigate patterns, and interpret results. Guided activities also introduce AI-assisted anomaly detection and encourage students to examine model outputs as evidence rather than automatic truth, including consideration of false positives, false negatives, and operational consequences.

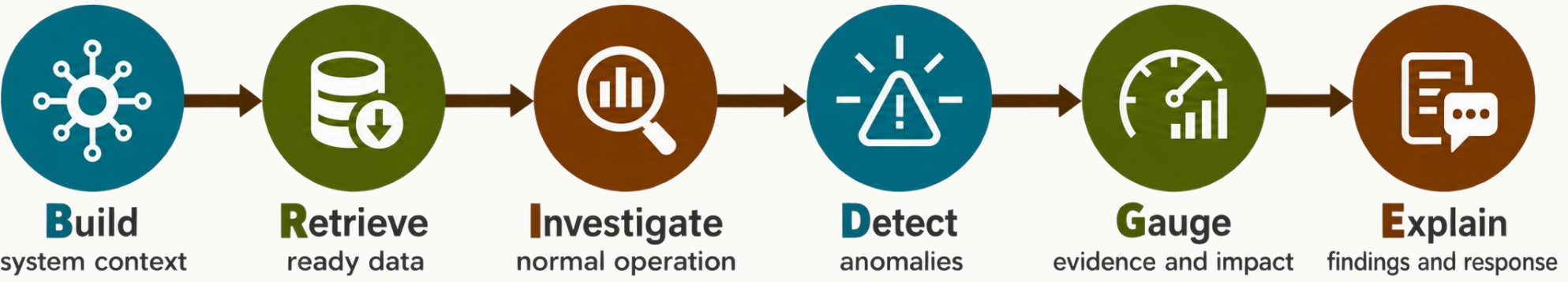
Students then transfer these skills to a revised final project. Each student develops a research question, selects and programs an Arduino-compatible sensor, collects a manageable original dataset, and analyzes and visualizes the data in a scaffolded Jupyter notebook. Students communicate their findings through a technical report that addresses methods, results, limitations, and potential cybersecurity or critical infrastructure applications.

The redesign introduces the **BRIDGE framework**, a structured process for helping students move from sensor data collection to analysis, anomaly detection, interpretation, and communication. It also emphasizes a progression from **awareness to guided practice to independent application**, helping beginning students see that the same fundamental process applies across scales.

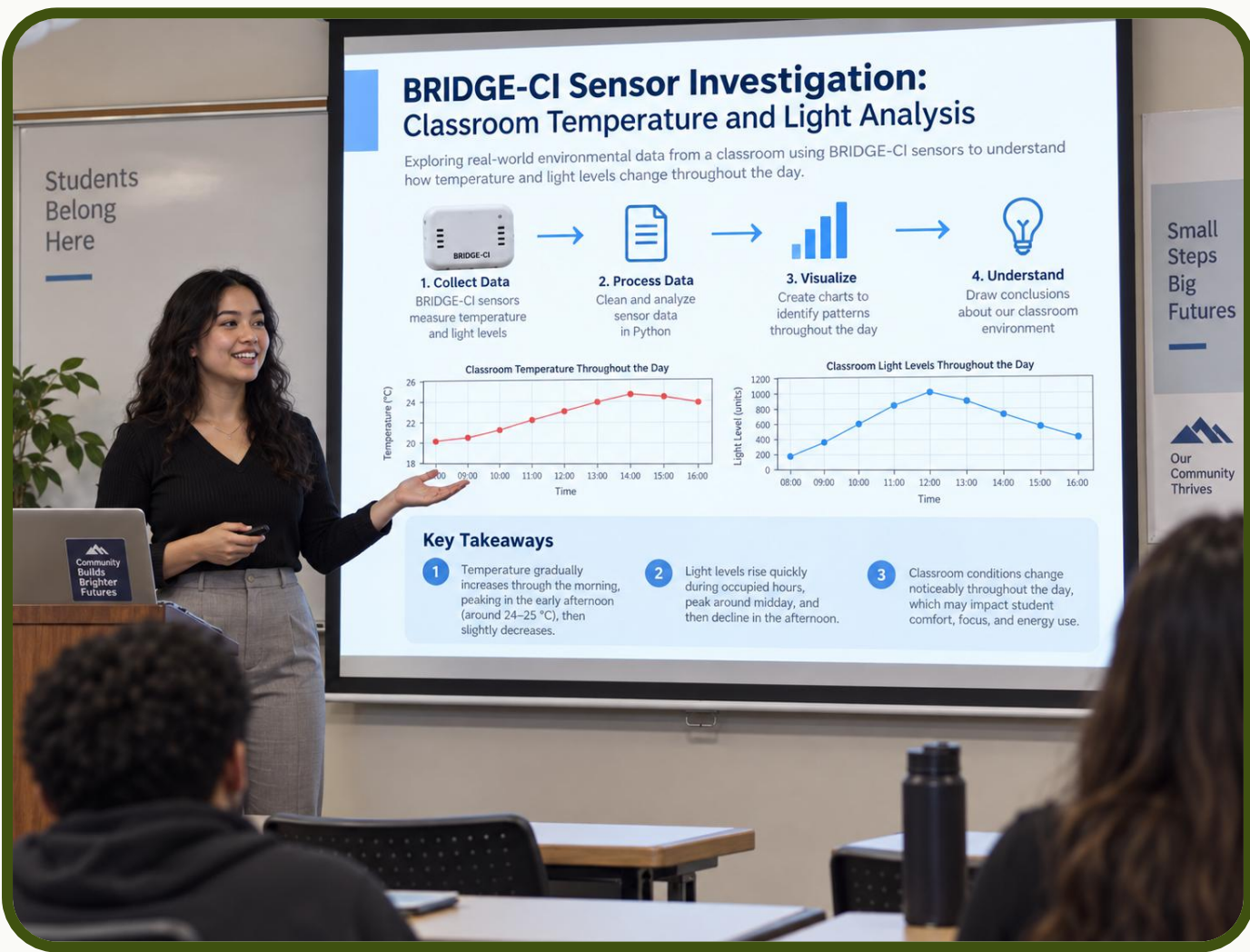
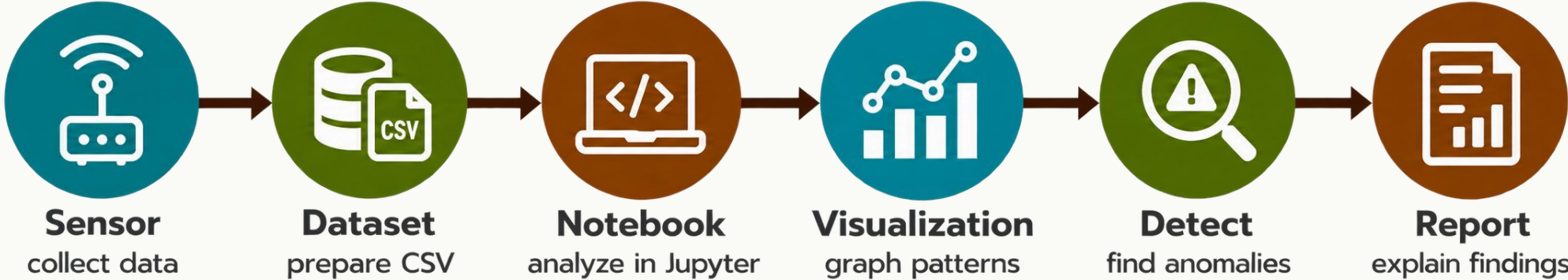
Ask a question → collect data → prepare data → visualize patterns → interpret results → communicate findings

FINAL PROJECT

The final project uses the **BRIDGE framework** to guide students through a complete sensor investigation, from developing a research question and collecting data to analysis, interpretation, and communication, culminating in a formal report and presentation.



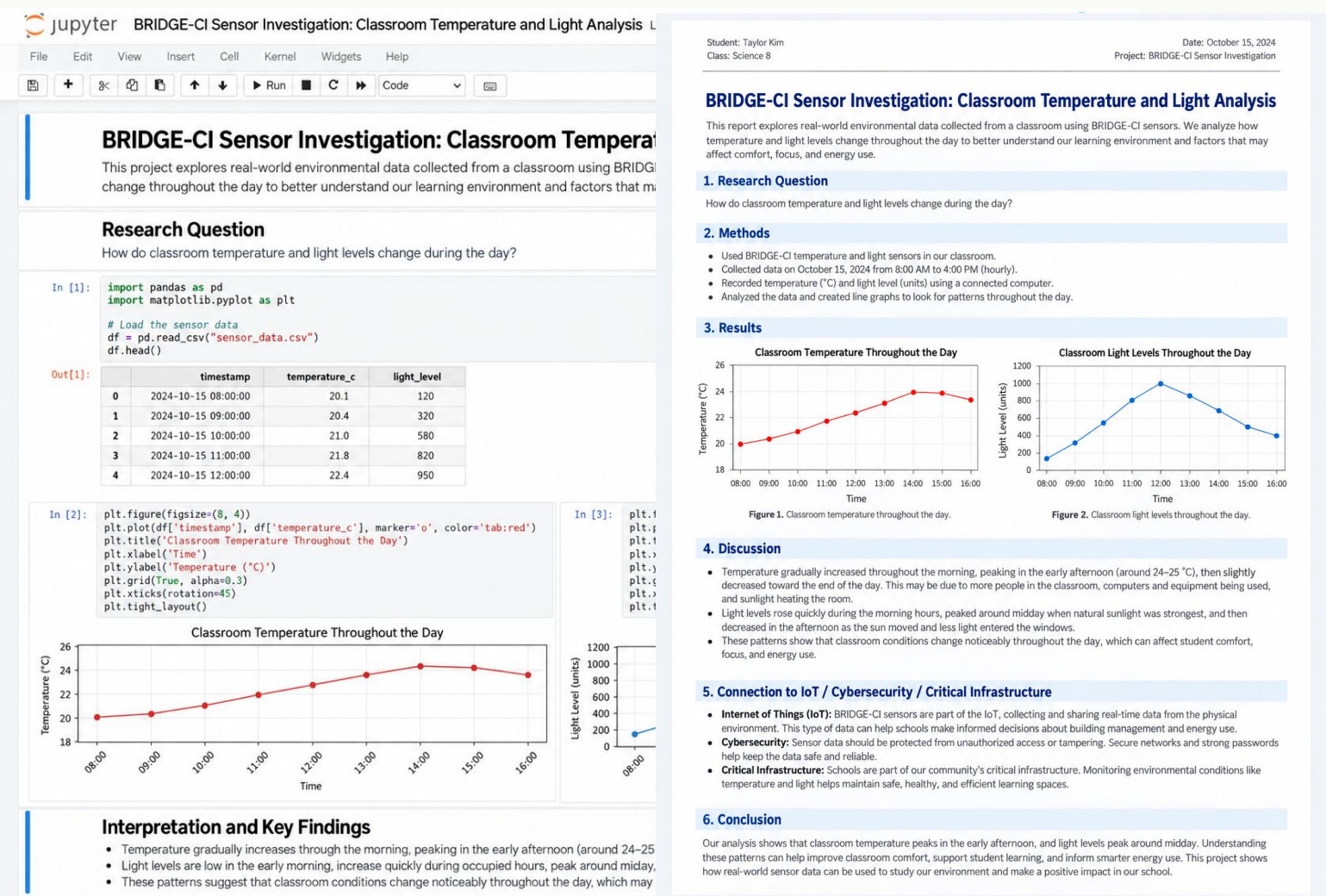
Students build a sensor system, collect and prepare data, analyze and visualize it in a Jupyter notebook, interpret patterns or anomalies, and communicate their findings in a technical report.



PROJECT GOALS

COLLECT & ANALYZE	INTERPRET & EVALUATE	CONNECT & COMMUNICATE
Gather sensor data with Arduino and explore it in Jupyter.	Identify patterns, anomalies, and possible cybersecurity or operational concerns.	Connect findings to scientific computing, cybersecurity, and critical infrastructure.

SCIENCE GATEWAY RESOURCES USED



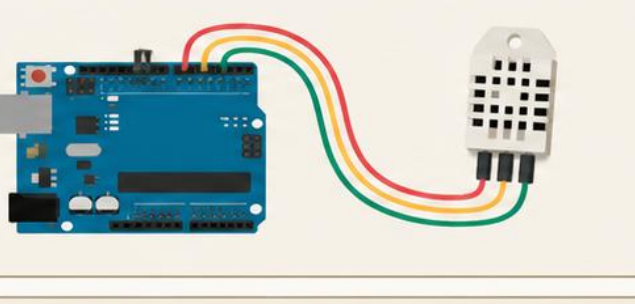
Resource	How It Is Used in LAN 120
Jetstream2	Introduces students to cloud-based research computing and resources beyond a local device.
Jupyter Notebooks	Used for guided data analysis, visualization, and interpretation of IoT sensor data.
Science Gateways	Introduces students to browser-based research tools, shared datasets, and computing resources.
Public Datasets	Provides authentic IoT, industrial, and cybersecurity data for guided analysis activities.
Data Visualization Tools	Helps students identify patterns, trends, thresholds, and anomalies in sensor data.
AI-Assisted Analysis	Introduces anomaly detection while emphasizing that students must interpret and verify results.

COURSE REDESIGN: BEFORE & AFTER

INITIAL SYLLABUS

Device-centered final project

- Build and program an IoT device.
- Demonstrate circuit and code functionality.
- Explain IoT hardware, sensors, networking, privacy, and security concepts.
- Limited emphasis on what happens to data after collection.



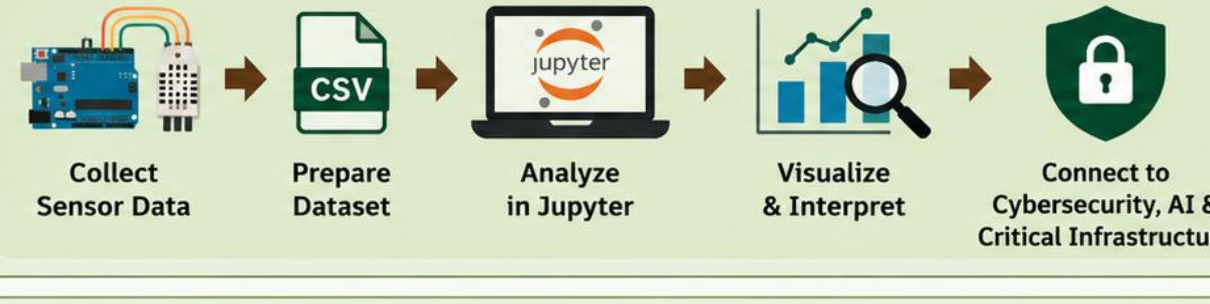
What Was Missing

- What the course covered before this redesign
- How students previously completed the equivalent project
- Limited access to data analysis tools and scientific computing workflows
- Less real-world relevance and connection to cybersecurity/critical infrastructure

MODIFIED SYLLABUS

Sensor-to-scientific-computing workflow

- Build and program an IoT device.
- Collect original sensor data.
- Analyze and visualize data in Jupyter.
- Interpret evidence and limitations.
- Connect findings to cybersecurity, AI-assisted analysis, and critical infrastructure.



What Was Added or Restructured

- Incorporated a sensor-to-scientific-computing workflow (BRIDGE-CI)
- Added Jupyter and data visualization analysis
- Connected findings to cybersecurity, AI, and critical infrastructure
- Expanded opportunities for real-world relevance and evidence-based analysis
- Assessment revised to reflect the new workflow and outcomes

IMPACT ON STUDENTS

- Broader access to computing resources** → Students gain exposure to Jupyter notebooks, Science Gateways, cloud/HPC resources, and authentic datasets.
- Stronger data skills** → Students move beyond building devices to collecting, visualizing, interpreting, and communicating sensor data.
- Greater confidence with advanced tools** → Scaffolded activities make AI, data analysis, and research computing more approachable for beginning students.
- Clearer real-world connections** → Students see how IoT data relates to cybersecurity, critical infrastructure, and scientific computing.

Lessons Learned

- What worked:** Scaffold the experience and focus on **data reasoning rather than advanced coding**.
- What I'd change:** Introduce tools and datasets in smaller steps to reduce setup and cognitive load.
- What surprised me:** How naturally a simple Arduino sensor project can connect to **Jupyter, AI, cybersecurity, and scientific computing**.

TARGET COURSE DESCRIPTION

Course Name	Internet of Things (IoT) Fundamentals
Course Number	LAN 120
Department	Computer Integrated Technologies
Enrollment	24 students each semester
Prerequisites	None
Project Repo	github.com/kristinechristensen/FacultyHack26

ABOUT THE COURSE

COURSE DESCRIPTION

IoT Fundamentals is an entry-level course for students with no prerequisites. It introduces electronics, Arduino programming, sensors, networking, IoT data, privacy, and security through hands-on work with connected devices. Students gain hands-on experience building and programming connected devices but traditionally have had limited exposure to what happens to sensor data after it is collected or how that data can be analyzed at larger scales.

The redesign was motivated by the **gap between students' hands-on experience with individual IoT devices and the larger data, computing, cybersecurity, and critical infrastructure environments in which those devices operate**. From *Sensors to Scientific Computing* extends the existing course by introducing Science Gateways, Jupyter notebooks, public datasets, cloud/HPC resources, and AI-assisted analysis in a beginner-friendly environment. Scaffolded activities allow students with no prior Python or HPC experience to explore authentic IoT, industrial, and cybersecurity data before applying the same workflow to data collected from their own sensors.

Course Goals

- IoT & Programming:** Build and program Arduino-based sensor systems and explain how sensor data moves from the device into a dataset for analysis.
- Electronics & Schematics:** Read basic schematics, identify electronic components, and construct Arduino-based circuits and sensor systems.
- Networking & Data:** Explain how IoT devices communicate and collect sensor data for analysis.
- Computational Thinking:** Use scaffolded Jupyter notebooks to load, inspect, visualize, and interpret IoT data, modifying selected variables and reasoning about patterns, thresholds, and anomalies.
- Professional Practice:** Document data collection methods, code, units, visualizations, interpretations, and limitations, and communicate findings clearly in a technical report.
- Science Gateways / HPC / AI:** Use Jetstream2, Science Gateways, Jupyter notebooks, public datasets, cloud/HPC resources, and AI-assisted anomaly detection to connect small-scale Arduino projects to larger scientific & industrial computing environments.

Key Content by Week

Week 1	Foundations + Gateway Awareness - Electronics, Arduino, sensors, and IoT overview - Introduction to Science Gateways, Jupyter notebooks, public datasets, and cloud/HPC resources
Week 2	Guided Notebook 1A - Open and navigate a Jupyter notebook - Run prepared code, load and inspect sensor data - Modify selected variables and explore the output
Week 3	Guided Notebook 1B - Create time-series visualizations - Explore rolling averages, thresholds, and patterns - Interpret results and draw conclusions
Week 4	Science Gateway Exploration - Use Science Gateways and Jetstream2 resources - Explore authentic IoT and other real-world datasets - Connect classroom sensor projects to larger data and computing environments
Week 5	AI-Assisted Analysis - Use a guided notebook for anomaly detection - Interpret results, including confusion matrix, false positives, and false negatives - Discuss potential applications and limitations
Week 6	Final Sensor Investigation (Part 1) - Develop a research question and select a sensor - Collect data using Arduino and sensors - Import data (e.g., CSV) into Jupyter for analysis
Week 7	Final Sensor Investigation (Part 2) - Analyze and visualize your own sensor data - Interpret findings and compare with expected results - Prepare a technical report and presentation
Week 8	Final Project Completion & Presentation - Finalize Jupyter analysis and technical report - Present your sensor investigation and findings - Complete the Arduino Certification Exam

Grade Computation

Assignments	65%
BRIDGE-CI Final Project*	20%
Arduino Certification Exam	15%

* **Note:** Students will not be graded on having the largest dataset or the most advanced code. They are graded on whether they can follow a sound process and explain evidence accurately.

References

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visit the course github repo!

PROJECT GITHUB REPO

github.com/kristinechristensen/FacultyHack26

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